ATOM COUNT CARD:
Sugar (Surcrose) in Bear Brand Fortified Milk



MOLE CONCEPT
ACTIVITY 02

A PRODUCT IDENTITY



Product: Bear Brand Fortified powdered milk drink, 33 g sachet.

Compound followed: sugars, treated as sucrose, $C_{12}H_{22}O_{11}$ (molecular, so counted in molecules)

Why it's in the product: natural milk sugar plus added sugar give sweetness and energy, as part of the 18.5 g carbohydrate per serving.

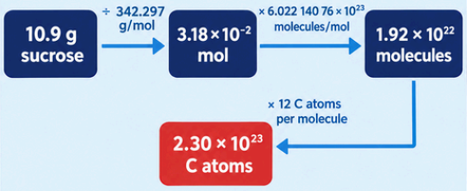
B BASE QUALITATIVE DATA

Quantity	Value	Source	Precision
Sugars per serving	10.9 g	Product label	**3 s.f. (limits)**
Serving size (powder)	33 g	Product label	2 s.f., not used in the main chain
Servings per sachet	1	Product label	Exact count
Formula of sucrose	$C_{12}H_{22}O_{11}$	PubChem (verify the CID, I believe 5988)	Exact subscripts
Atomic masses	C 12.011, H 1.008, O 15.999 g/mol	CIAAW (2024)	Not limiting
Avogadro constant	$6.022\,140\,76 \times 10^{23}$ mol^{-1}	BIPM (2019)	Exact

C MOLE MASS BUILD-UP

Element	Atoms	g/mol each	Contribution (g/mol)	% by mass
C	12	12.011	144.132	42.107
H	22	1.008	22.176	6.479
O	11	15.999	175.989	51.414
$C_{12}H_{22}O_{11}$	**45**		**342.297**	**100**

D THE MOLE MAP



Grams never convert straight to particles.
Always pass through moles, and name the entity.

E MOLE CONVERSION CALCULATIONS

1 Molar mass

$12(12.011) + 22(1.008) + 11(15.999) = 144.132 + 22.176 + 175.989$
$= 342.297$ g/mol

2 Label bridge

$1 \text{ sachet} \times \frac{1 \text{ serving}}{1 \text{ sachet}} \times \frac{10.9 \text{ g sugars}}{1 \text{ serving}}$
$= 10.9$ g sucrose

3 Mass → moles

$10.9 \text{ g} \times \frac{1 \text{ mol}}{342.297 \text{ g}} = 0.0318437 \text{ mol} \rightarrow 3.18 \times 10^{-2}$ mol sucrose

4 Moles → molecules

$0.0318437 \text{ mol} \times \frac{6.022\,140\,76 \times 10^{23} \text{ molecules}}{1 \text{ mol}} = 1.91767 \times 10^{22}$
$\rightarrow 1.92 \times 10^{22}$ sucrose molecules

5 Carbon atoms

$1.91767 \times 10^{22} \text{ molecules} \times \frac{12 \text{ C atoms}}{1 \text{ molecule}} = 2.30121 \times 10^{23}$
$\rightarrow 2.30 \times 10^{23}$ C atoms

6 Percent composition + two-route check (carbon)

$\%C = \frac{12(12.011)}{342.297} \times 100\% = 42.107\%$

Route A: $10.9 \text{ g} \times 0.42107 = 4.5897$ g C.

Route B: $0.0318437 \text{ mol} \times \frac{12 \text{ mol C}}{1 \text{ mol sucrose}} \times \frac{12.011 \text{ g C}}{1 \text{ mol C}} = 4.5897$ g C.

Both routes → 4.59 g C ✓ agree

7 Reverse to formula

per 100 g, C 3.5057 mol, H 6.4272 mol, O 3.2135 mol;
÷ 3.2135 → C 1.091 : H 2.000 : O 1.000; ×11 →
C 12 : H 22 : O 11 → $C_{12}H_{22}O_{11}$.
n = 342.297 ÷ 342.297 = 1 (empirical = molecular)

8 Scale-up and reverse

200 sachets × 0.0318437 mol/sachet = 6.3687 mol → 6.37 mol sucrose
(2.18×10^3 g; 3.84×10^{24} molecules).

Reverse: $1.00 \text{ mol} \times \frac{342.297 \text{ g}}{1 \text{ mol}} \times \frac{1 \text{ sachet}}{10.9 \text{ g}} = 31.403 \rightarrow 3.14 \times 10^1$ sachets

F SCALE-UP: FROM ONE SACHET TO THE WHOLE SCHOOL

1 One sachet (given):
3.18×10^{-2} mol
(0.0318 mol)
This is the amount of sucrose in 1 sachet of the product.
(From the molar mass and mass given in the card: 10.9 g sucrose per sachet)

2 Number of sachets to make a whole school:
Total mol needed (school) = 3.18×10^{-2} mol/sachet × 200 sachets
= **6.36 mol**

3 Convert moles to grams (carbon atoms):
Mass = moles × molar mass
= 6.36 mol × 12.011 g/mol
= **76.36 g C**
12.011 g/mol is the molar mass of carbon (C). (From the periodic table)

4 Reverse to get the formula:
Per 100 g: : C 3.5057 mol, H 6.4272 mol, O 3.2135 mol.
Divide by 3.2135 (oxygen):
C 1.091 : H 2.000 : O 1.000
(≈ 1 : 2 : 1)
This gives the empirical formula: $C_1H_2O_1$ (which is the same as $C_1H_2O_1$)

5 Find the molar mass of the compound:
$C_1H_2O_1$ = (12.011 × 1) + (1.008 × 2) + (15.999 × 1)
= 12.011 + 2.016 + 15.999
= **30.026 g/mol**
This is the molar mass of sucrose ($C_1H_2O_1$). (≈ 30.03 g/mol)

6 Scale-up and reverse:
200 sachets × 0.0318437 mol/sachet = 6.3687 mol
→ **6.37 mol sucrose** (2.18×10^{24} molecules).

7 FINAL ANSWER:
6.37 mol sucrose
(2.18×10^{24} molecules)

G SIGNIFICANT FIGURE CHECK

Calculator: $1.917\,671\,5 \times 10^{22}$ molecules
Reported: 1.92×10^{22} molecules
Limiting value: "10.9 g" (3 s.f.). Atomic masses and percentages are not limiting.
Exact: N_a, subscripts, sachet counts, 1 g = 1000 mg.
Note on the label: nutrition labels round values, so the true sugar content may differ slightly.

ONE-NUMBER TAKEAWAY

One sachet holds about 1.92×10^{22} sugar molecules, yet it takes the sugar from about 31 sachets to reach a single mole. A mole is an enormous count of molecules, but one mole of sucrose weighs only 342 g.

H REFERENCES

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Commission on Isotopic Abundances and Atomic Weights. (2024). Abridged standard atomic weights. International Union of Pure and Applied Chemistry. https://www.ciaaw.org/abridged-atomic-weights.htm

National Center for Biotechnology Information. (n.d.). PubChem compound summary for CID 5988, sucrose. Retrieved October 18, 2026, from https://pubchem.ncbi.nlm.nih.gov/compound/5988